

Thermodynamic State Vector Interface and Monad Process Architecture in the Web of Life

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Abstract

Planetary-scale ecological models frequently struggle to reconcile empirical biogeochemical cycles with the rigorous dictates of non-equilibrium thermodynamics. In Sprint 015, we establish the formal TypeScript interface contracts and state-transforming monad architecture for the **Web of Life** engine (`src/thermodynamics/types.ts` and `src/thermodynamics/methods.ts`). By embedding exact mathematical representations of the First Law of Thermodynamics (energy conservation) and the Second Law of Thermodynamics (Clausius-Duhem entropy generation inequality and the Gouy-Stodola exergy destruction theorem), our framework models open biogeochemical subsystems (Carbon, Nitrogen, Phosphorus, and Water) as rigorous thermodynamic control volumes. We demonstrate how boundary flux arrays, internal entropy generation rates ($\dot{S}_{\text{gen}}$), and exergy destruction rates ($\dot{I} = T_0 \dot{S}_{\text{gen}}$) are enforced programmatically to guarantee thermodynamic consistency across macro-ecological simulations.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

1 Introduction and Systems Ecology Framing

In systems ecology, ecosystems and planetary nutrient cycles are fundamentally open thermodynamic engines driven by external solar irradiance and geothermal heat. Traditional compartment models track mass and energy balances but often omit entropy tracking, leading to unphysical configurations where energy or entropy constraints are violated.

The **Web of Life** framework addresses this by treating every ecological subsystem k as a non-equilibrium thermodynamic system governed by strict conservation laws and irreversible entropy production. Sprint 015 formalizes this foundation by introducing concrete interface contracts and monad transformation pipelines that enforce physical feasibility at every simulation step.

2 Core Mathematical Formalism

2.1 First Law: Energy Conservation

For any open subsystem k , the time rate of change of total internal energy E_k is governed by:

$$\frac{dE_k}{dt} = \dot{Q}_k - \dot{W}_k + \sum_{in} \dot{m}_{in} h_{in} - \sum_{out} \dot{m}_{out} h_{out} \quad (1)$$

Where $\dot{Q}_k$ is the net heat transfer rate, $\dot{W}_k$ is the work rate, and $\dot{m}h$ represents convective enthalpy fluxes across subsystem boundaries. First law closure is verified via residual calculations:

$$\text{Residual} = \left| E_k(t + \Delta t) - E_k(t) - \int_0^{\Delta t} \left(\dot{Q}_k - \dot{W}_k + \sum \dot{m}h \right) dt \right| < 10^{-10} \text{ J} \quad (2)$$

2.2 Second Law: Entropy Generation and Exergy Destruction

The entropy balance of subsystem k is given by:

$$\frac{dS_k}{dt} = \sum_j \frac{\dot{Q}_{kj}}{T_j} + \sum_{in} \dot{m}_{in} s_{in} - \sum_{out} \dot{m}_{out} s_{out} + \dot{S}_{gen,k} \quad (3)$$

The Clausius-Duhem inequality mandates that internal entropy generation cannot be negative:

$$\dot{S}_{gen,k} \geq 0 \quad (4)$$

Using the Gouy-Stodola theorem, the rate of exergy destruction ($\dot{I}_k$), representing lost thermodynamic work potential at ambient dead-state temperature T_0 , is computed as:

$$\dot{I}_k = T_0 \dot{S}_{gen,k} \geq 0 \quad (5)$$

3 Implementation Architecture

The thermodynamic engine is structured around three primary TypeScript artifacts:

1. **ThermodynamicStateVector**: Encapsulates instantaneous temperature, dead-state temperature (T_0), internal energy, total entropy, entropy generation rate ($\dot{S}_{gen}$), exergy destruction rate ($\dot{I}$), and boundary flux arrays.
2. **IThermodynamicModel**: Interface implemented by **BaseCycle** (extending to carbon, nitrogen, phosphorus, and water cycles) ensuring method signatures for **stepThermodynamics(dt)**, **validateFirstLaw()**, and **validateSecondLaw()**.
3. **stepThermodynamicMonad**: Pure functional state transformer executing discrete time-step integration of mass-energy boundary fluxes followed by projection guard enforcement of $\dot{S}_{gen} \geq 0$.

4 Verification and Conclusion

Sprint 015 successfully establishes absolute mathematical grounding for the Web of Life simulation environment. Unit tests confirm strict adherence to energy conservation and non-negative entropy generation across all macro-ecological cycles.

For full source code, architecture specifications, and database schemas, please visit the official repository at <https://github.com/pascalranoroarijaona/WebOfLife>.